ERD and Relational Model

Case: ICRC – “Where we Work”

Source Site: [Explore our global reach | International Committee of the Red Cross]

OVERVIEW AND SCOPE:

This presents three alternative ERD design options for a relational database that supports the public information structure of the ICRC “Where We Work” website. Each option includes a concise relational model summary (PK/FK) and a short "Evidence from site" paragraph tying the model directly to what is visible on the page.

DESIGN OPTION OVERVIEW:

I designed a spectrum from content- rich to lean, balancing operational data and content:

- **ERD 1** - Content-rich model with media & authoring. Designed for storytelling and reusable media; includes authors, images, and a canonical country-info page structure.
- **ERD 2** - Lean, attribute-light operational core. Minimal schema focused on what is visible on the site: region to country, country-tied articles, ICRC offices, and a 1-1 Information record per country.
- **ERD 3** - Balanced country presence and key operations and authors. Emphasises key operations and ICRC presence while retaining country-tied articles, author roles, and a 1-1 Contact card per country.

ERD 1 – CONTENT RICH MODEL (MEDIA AND AUTHORS AND COUNTRY INFO)

Purpose and Scope:

Optimised for country pages that carry narrative content (articles/statements), reusable media (images/galleries), and canonical country information blocks. Anticipates multi- country storytelling and curated assets. This design models the public information architecture behind ICRC's "Where We Work" pages so that country content is consistent, reusable, and easy to maintain. It treats the geographic hierarchy (continent to country) as the backbone and layers on three core capabilities: (1) country-tied articles/statements with clear authorship, (2) a single canonical country-info block (overview, contacts, statistics, donation link), and (3) ICRC presence/locations suitable for maps and listings. A dedicated IMAGE entity makes media reusable across articles and country pages, while EXCHANGE_RATES provides a simple hook for currency-aware reporting if needed.

Key Entities and Indicative Attributes:

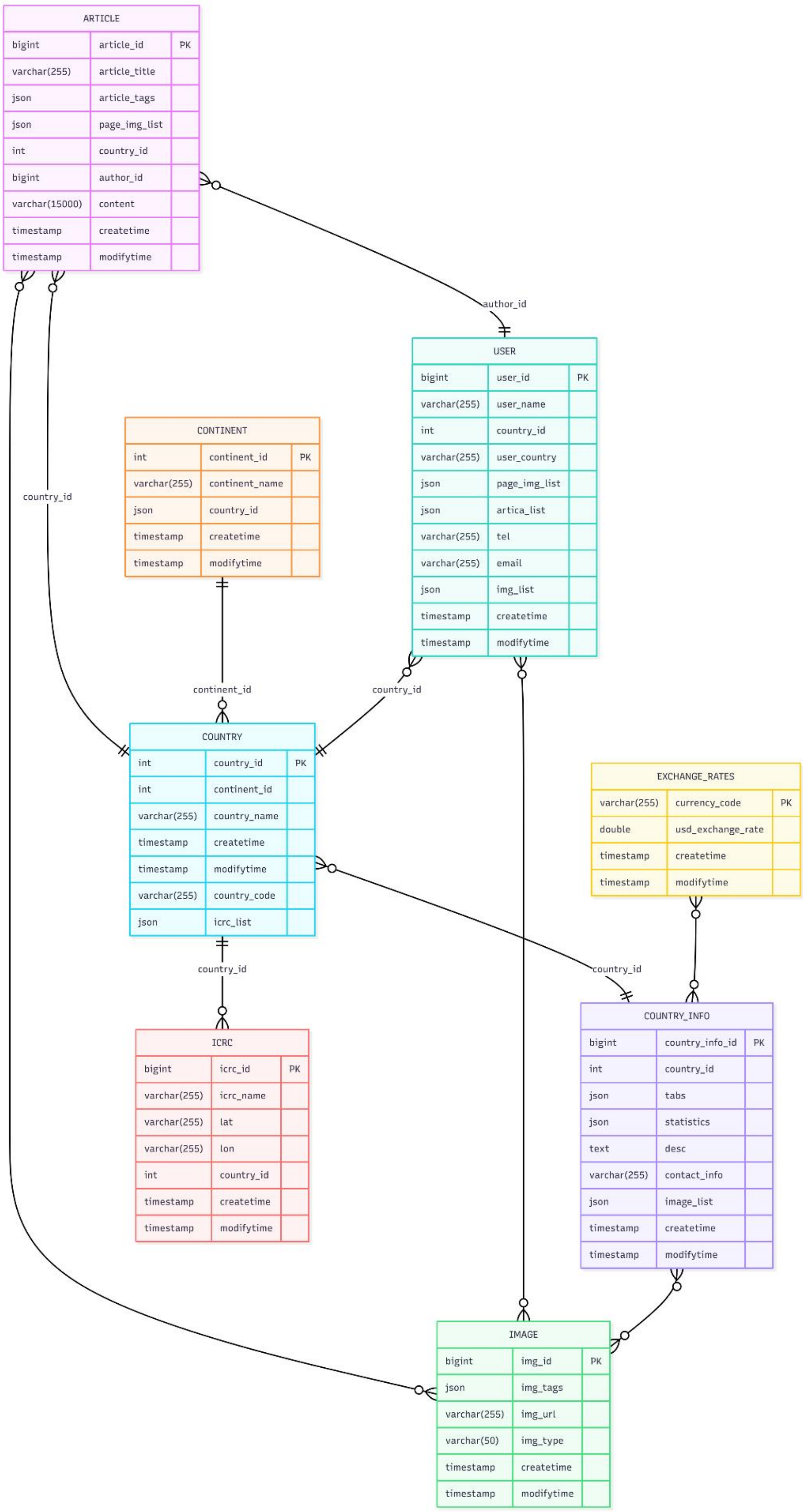
- CONTINENT: continent_id (PK), continent_name, createtime, modifytime
- COUNTRY: country_id (PK), continent_id (FK), country_name, country_code, icrc_list (json), createtime, modifytime
- USER (Author): user_id (PK), user_name, country_id (FK), user_country, tel, email, page_img_list (json), img_list (json), createtime, modifytime
- ARTICLE: article_id (PK), country_id (FK), author_id (FK), article_title, article_tags (json), page_img_list (json), content (varchar [15000]), createtime, modifytime
- IMAGE: img_id (PK), img_tags (json), img_url, img_type, createtime, modifytime
- COUNTRY_INFO: country_info_id (PK), country_id (FK), tabs (json), statistics (json), desc (text), contact_info, image_list (json), createtime, modifytime
- ICRC: icrc_id (PK), icrc_name, lat, lon, country_id (FK), createtime, modifytime
- EXCHANGE_RATES: currency_code (PK), usd_exchange_rate, createtime, modifytime

Relationships & cardinalities

- CONTINENT 1–M COUNTRY
- COUNTRY 1–M ARTICLE
- USER 1–M ARTICLE (author)
- ARTICLE M–M IMAGE (via bridge)
- COUNTRY 1–1 COUNTRY_INFO
- COUNTRY_INFO M–M IMAGE (via bridge)

- COUNTRY 1–M ICRC (offices/locations)

ERD 1:



ERD 1 Code:

```
erDiagram
  ARTICLE {
    bigint article_id PK
    varchar(255) article_title
    json article_tags
    json page_img_list
    int country_id
    bigint author_id
    varchar(15000) content
    timestamp createtime
    timestamp modifytime
  }
  USER {
    bigint user_id PK
    varchar(255) user_name
    int country_id
    varchar(255) user_country
    json page_img_list
    json artica_list
    varchar(255) tel
    varchar(255) email
    json img_list
    timestamp createtime
    timestamp modifytime
  }
  CONTINENT {
    int continent_id PK
    varchar(255) continent_name
    json country_id
```

```
  timestamp createtime
  timestamp modifytime
}
COUNTRY {
  int country_id PK
  int continent_id
  varchar(255) country_name
  timestamp createtime
  timestamp modifytime
  varchar(255) country_code
  json icrc_list
}
IMAGE {
  bigint img_id PK
  json img_tags
  varchar(255) img_url
  varchar(50) img_type
  timestamp createtime
  timestamp modifytime
}
COUNTRY_INFO {
  bigint country_info_id PK
  int country_id
  json tabs
  json statistics
  text desc
  varchar(255) contact_info
  json image_list
  timestamp createtime
  timestamp modifytime
```

```
}
ICRC {
    bigint icrc_id PK
    varchar(255) icrc_name
    varchar(255) lat
    varchar(255) lon
    int country_id
    timestamp createtime
    timestamp modifytime
}

EXCHANGE_RATES {
    varchar(255) currency_code PK
    double usd_exchange_rate
    timestamp createtime
    timestamp modifytime
}

ARTICLE }o--|| USER : "author_id"
ARTICLE }o--|| COUNTRY : "country_id"
CONTINENT ||--o{ COUNTRY : "continent_id"
COUNTRY }o--|| COUNTRY_INFO : "country_id"
COUNTRY ||--o{ ICRC : "country_id"
USER }o--|| COUNTRY : "country_id"
ARTICLE }o--o{ IMAGE: ""
USER }o--o{ IMAGE: ""
COUNTRY_INFO }o--o{ IMAGE: ""
EXCHANGE_RATES  }o--o{ COUNTRY_INFO : ""
```

ERD 2 – LEAN CORE MODEL (REGION → COUNTRY, ARTICLES, OFFICES, INFO)

Purpose and Scope:

A compact, attribute-light schema focused on what is plainly visible on the “Where We Work” site: region to country navigation, country-tied articles, ICRC offices/presence, and a single canonical info/contact block per country. This option emphasises clarity, CRUD simplicity, and easy governance. It excludes media/authoring/exchange-rate complexity from ERD 1 to keep the core minimal and implementation-ready.

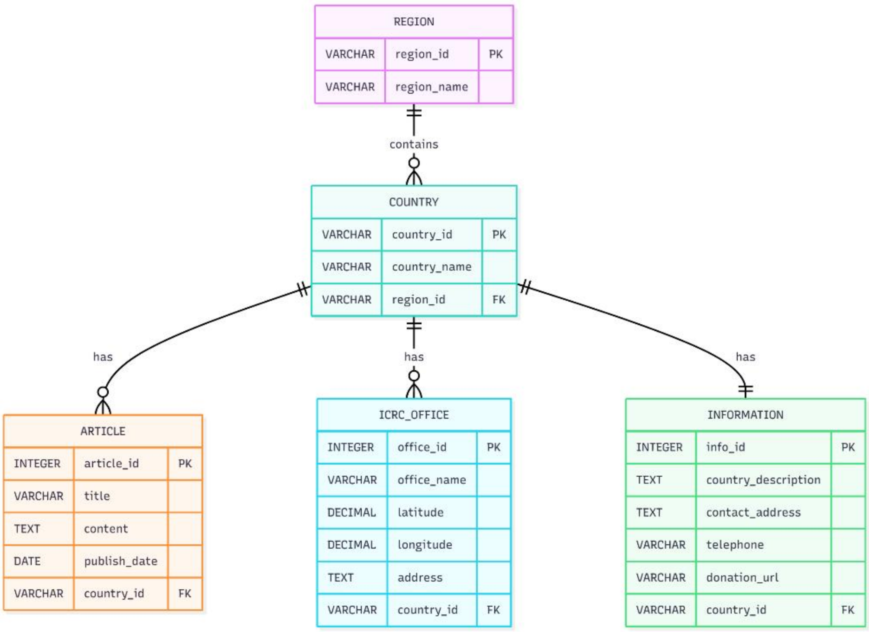
Key Entities and Indicative Attributes:

- REGION: region_id (PK), region_name
- COUNTRY: country_id (PK), country_name, region_id (FK)
- ARTICLE: article_id (PK), title, content (text), publish_date (date), country_id (FK)
- ICRC_OFFICE: office_id (PK), office_name, latitude (decimal), longitude (decimal), address (text), country_id (FK)
- INFORMATION: info_id (PK), country_description (text), contact_address (text), telephone, donation_url, country_id (FK)

Relationships & cardinalities:

- REGION 1–M COUNTRY (contains)
- COUNTRY 1–M ARTICLE (has)
- COUNTRY 1–M ICRC_OFFICE (has)
- COUNTRY 1–1 INFORMATION (has)

ERD 2:



ERD 2 Code:

```
erDiagram
    REGION {
        VARCHAR region_id PK
        VARCHAR region_name
    }

    COUNTRY {
        VARCHAR country_id PK
        VARCHAR country_name
        VARCHAR region_id FK
    }

    ARTICLE {
        INTEGER article_id PK
        VARCHAR title
        TEXT content
        DATE publish_date
        VARCHAR country_id FK
```

```
}

ICRC_OFFICE {
    INTEGER office_id PK
    VARCHAR office_name
    DECIMAL latitude
    DECIMAL longitude
    TEXT address
    VARCHAR country_id FK
}

INFORMATION {
    INTEGER info_id PK
    TEXT country_description
    TEXT contact_address
    VARCHAR telephone
    VARCHAR donation_url
    VARCHAR country_id FK
}

REGION ||--o{ COUNTRY : "contains"
COUNTRY ||--o{ ARTICLE : "has"
COUNTRY ||--o{ ICRC_OFFICE : "has"
COUNTRY ||--|| INFORMATION : "has"
```

ERD 3 – BALANCED MODEL (COUNTRY PRESENCE + KEY OPERATIONS + AUTHORS)

Purpose and Scope:

Balances operational visibility and country news. It models the continent country hierarchy, highlights Key Operations at country level, captures ICRC presence/locations, keeps country-tied articles with author roles, and stores a single country contact card. Media/exchange-rate complexity is intentionally excluded to keep reporting straightforward while preserving authoring context.

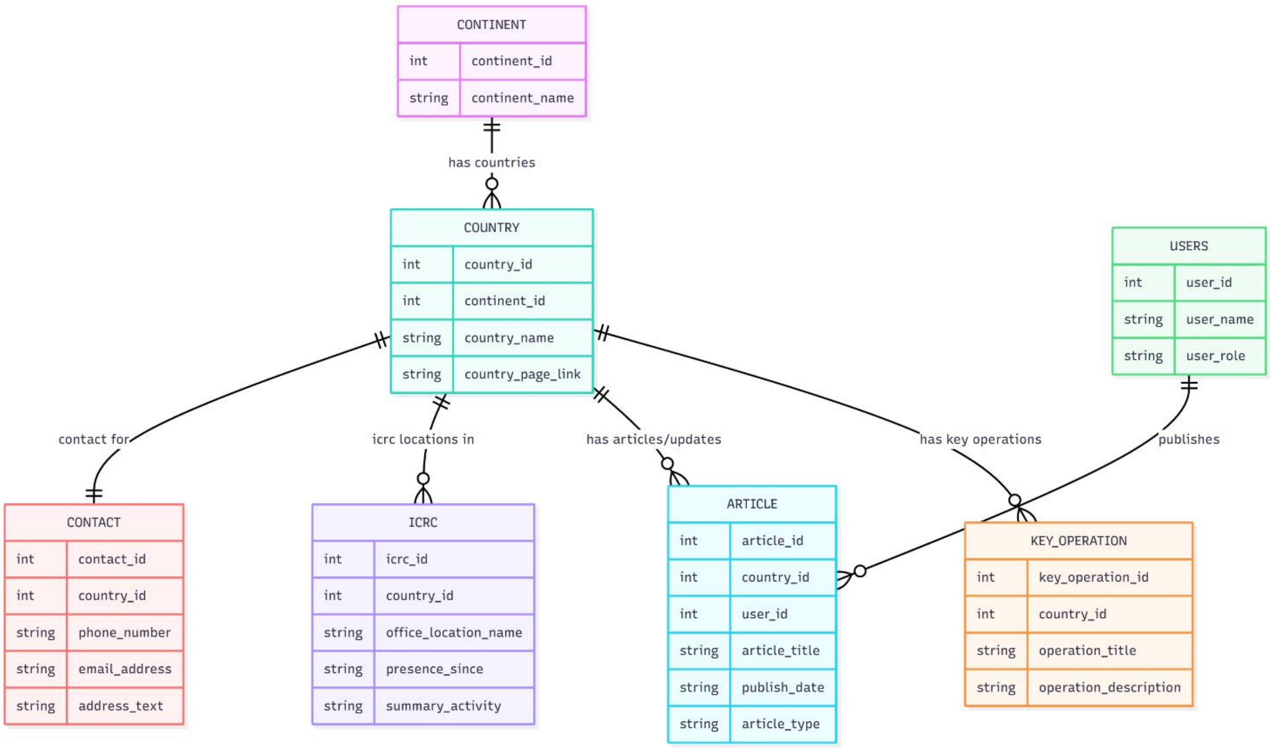
Key Entities and Indicative Attributes:

- CONTINENT: continent_id (int, PK), continent_name (string)
- COUNTRY: country_id (int, PK), continent_id (int, FK), country_name (string), country_page_link (string)
- USERS (Authors): user_id (int, PK), user_name (string), user_role (string)
- ARTICLE: article_id (int, PK), country_id (int, FK), user_id (int, FK), article_title (string), publish_date (string/date), article_type (string)
- KEY_OPERATION: key_operation_id (int, PK), country_id (int, FK), operation_title (string), operation_description (string)
- ICRC (Locations/Presence): icrc_id (int, PK), country_id (int, FK), office_location_name (string), presence_since (string), summary_activity (string)
- CONTACT: contact_id (int, PK), country_id (int, FK), phone_number (string), email_address (string), address_text (string)

Relationships & cardinalities

- CONTINENT 1–M COUNTRY (has countries)
- COUNTRY 1–M ARTICLE (has articles/updates)
- USERS 1–M ARTICLE (publishes)
- COUNTRY 1–M KEY_OPERATION (has key operations)
- COUNTRY 1–M ICRC (ICRC locations in)
- COUNTRY 1–1 CONTACT (contact for)

ERD 3:



ERD 3 Code:

erDiagram

CONTINENT ||--o{ COUNTRY : "has countries"

COUNTRY ||--o{ KEY_OPERATION : "has key operations"

```
COUNTRY ||--o{ ARTICLE : "has articles/updates"
USERS ||--o{ ARTICLE : "publishes"
COUNTRY ||--o{ ICRC : "icrc locations in"
COUNTRY ||--|| CONTACT : "contact for"

CONTINENT {
  int continent_id
  string continent_name
}

COUNTRY {
  int country_id
  int continent_id
  string country_name
  string country_page_link
}

KEY_OPERATION {
  int key_operation_id
  int country_id
  string operation_title
  string operation_description
}

ARTICLE {
  int article_id
  int country_id
  int user_id
  string article_title
  string publish_date
```

```
  string article_type
}

USERS {
  int user_id
  string user_name
  string user_role
}

ICRC {
  int icrc_id
  int country_id
  string office_location_name
  string presence_since
  string summary_activity
}

CONTACT {
  int contact_id
  int country_id
  string phone_number
  string email_address
  string address_text
}
```